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Research Article

Visual perception-informed urban design toolkit: Computational urban morphology optimisation to inform real-time perceived safety

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ABSTRACT

The perceived safety of street scenes significantly affects the travel behaviors and social interactions of urban dwellers, particularly females, which ultimately matters to the inclusiveness of cities. Despite efforts to utilize street view imagery (SVI) for auditing perceived safety in urban areas, most studies remain analytical, with limited integration into urban design's form-based ideation workflow. The reason is mechanism dependency: few designers use both analytics (Python) and form-based design (Rhino) tools fluently. The outcome should not be overlooked: when the humanistic pedestrian experience cannot be explicitly integrated into the design process, the planning results may further marginalise disadvantaged groups. To bridge the gap between urban analytics and design, we propose a computational framework that automates the evaluation of pedestrian-oriented perceived safety in real-time, linking form-based urban design features, particularly greenery, to visual safety perception within urban canyons. By integrating datasets such as Place Pulse 2.0 and environmental attributes, the perceived safety of different urban canyons with varying urban forms and streetscape configurations can be seamlessly updated using machine learning in Rhino Grasshopper. Our tests show that in areas with the same urban density and height-width ratio, specific greenery configurations, such as tree density and green coverage, significantly improve perceived safety. Subsequently, urban canyon forms are categorised based on perceived safety outcomes to provide urban design guidelines. Notably, real-time visualisations from Stable Diffusion are further incorporated to improve the Rhino-based framework's usability in real life. This computational framework integrates visual perceptions into form-based urban form ideation (especially greenery characteristics): it alleviates the politics of difference in urban design practice, supporting the facilitation of more inclusive public spaces.

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1. Introduction

1.1. Background

In the context of accelerated urbanisation, the quality of urban public spaces and residents' daily experiences have become key concerns in urban design (Maas et al., 2009; Ye et al., 2025). Objective safety, among many indicators, is typically measured by quantifiable metrics such as crime rates, traffic accident records, and surveillance data, and has been widely evaluated to improve urban environments (Cozens & Love, 2015; Ewing & Handy, 2009; Jeffery, 1971; Newman, 1978). Nevertheless, such objective proxies do not fully capture residents' everyday experience (Sreetheran & Van Den Bosch, 2014), with (subjectively) perceived safety remaining poorly studied (Kim et al., 2024).

The subjective measure offers an alternative approach, emphasising individual safety perception that judges the local neighbourhood environment (Liu et al., 2024; Mehta, 2014; Ye et al., 2025). Notably, the perceived safety is influenced by visual, psychological, and environmental factors (Lis et al., 2019; Mehta, 2014; Mouratidis, 2019; Qiu et al., 2023), and in turn can shape multifaceted urban phenomena, including active travel (Song, Huang, et al., 2024; Wei et al., 2025; Wu et al., 2023), economic activities (Qiu et al., 2022), social interactions and emotional connections (Jacobs, 1961; Zhao et al., 2024), ultimately informing how public spaces are used (Campos Ferreira et al., 2022).

However, perceived safety assessments have yet to be effectively integrated into the urban design process to facilitate human-centred perspectives. When such humanistic pedestrian experience cannot be explicitly incorporated during urban design's decision-making feedback loop, the outcome of city planning could derive biased projections on the environment that can further disadvantage pedestrians in everyday travel, especially vulnerable groups such as females (Basu et al., 2023; Mark & Heinrichs, 2019). This reflects a failure to recognise and respond to the diverse needs of different social groups in policy-making, which is a core concern discussed in the politics of difference (Smeds et al., 2020). Existing urban design often prioritises traffic flow for efficiency, and urban policies have shown renewed interest in emphasising pedestrian-oriented safety for a better quality of life and well-being through human-centred urban design (UN Habitat, 2019).

Perceived safety has been reported to be shaped by a combination of environmental characteristics, including visual features (e.g., sky, building, street furniture), land use, and urban form and layout (Van Veghel et al., 2024). Among these, greenery design plays a critical role in influencing spatial experience at the human scale (Lis et al., 2019; Qiu et al., 2022; Su et al., 2023) across the day and night (Ye et al., 2025). However, the direction and mechanism of the relationship remain mixed. Some studies show that greenery, particularly tree canopy size and height, positively impacts perceived safety (Basu et al., 2023; Li et al., 2015), while several other studies reported the negative influence of greenery (Jansson et al., 2013; Jorgensen et al., 2002; Maruthaveeran, 2016; Nasar & Fisher, 1993). Arguably, the discrepancy is primarily attributed to the way green form is measured, often relying on green pixel ratios in images without accounting for spatial and morphological features.

While existing over-simplified quantity-focused greenery analysis frameworks are efficient for analysing basic spatial configurations, they often fail to integrate further detailed greenery metrics, such as canopy size, trunk girth, or tree height, with perceptual data. These characteristics can better reflect the tree form but remain poorly understood. This ill-addressed aspect not only affects the distribution of light and shadow on streets but also psychologically influences safety through the concealment effect (Kuo et al., 1998). This disconnection leaves a critical gap in understanding how specific greenery characteristics interact with urban form to influence perceived safety, directly affecting urban planting selection in real practice. These complexities highlight the urgent need to better understand how street greenery characteristics affect safety perception across diverse urban settings, to guide better context-sensitive urban design strategies (Sreetheran & Van Den Bosch, 2014).

Moreover, while the importance of human-centred urban design in enhancing perceived safety has been increasingly recognised (Kim et al., 2024), a critical limitation in existing studies is their reliance on site-specific explanatory analyses, which often fail to produce generalisable forward-looking insights for urban design in diverse urban contexts. While localised case analyses offer rich contextual reflection details, their findings are usually constrained by unique sociocultural, regulatory, or morphological factors (Kang & Kang, 2016; Mouratidis, 2019). This impedes the translation of research findings into scalable design guidelines directly applicable to cities with varying densities, geometries, and socioeconomic conditions.

In light of this, developing parametric urban canyon models within a speculative framework defined by parametric abstractions of street morphology (e.g., height-to-width ratios, tree density) has emerged as a robust framework for isolating the impact of spatial configurations on human perceptions (Van Veghel et al., 2024). However, most prior studies focused on individual environmental features in isolation rather than their combined influence within real-world design workflows (Ewing & Handy, 2009; Marquet & Miralles-Guasch, 2015; Van Rijswijk & Haans, 2018). How to develop a scenario-based parametric predictive model, serving as a surrogate to estimate perceptual outcomes while jointly manipulating tree layout, form, street width, building spacing, and lighting, integrating these variables into early-stage urban design processes remains largely underexplored.

1.2. Aims and contribution

To address these research gaps, this study proposes to explore how street greenery characteristics affect perceived safety and develop a scalable computational design framework that can optimise perceived safety using parametric tools and Machine Learning (ML). Specifically, the study consists of four research objectives. First, it aims to reveal the poorly understood relationship between street trees' various attributes (e.g., height, age, girth size) and perceived safety. Second, the research develops a parametric framework that integrates urban canyon geometry (e.g., height-to-width ratio) and vegetation metrics to assess perceived safety in urban spaces. Third, data-driven strategies are identified using the parametric framework to optimise greenery design for enhancing perceived safety. Lastly, a real-time generative Artificial Intelligence (AI) module is developed that integrates perception-based visual simulation into parametric workflows, guiding greenery and safety decisions across varied contexts, effectively bridging the long-standing gap between urban analytics and design.

This study contributes to the planning and urban design literature by empirically analysing tree features and leveraging ML-based simulations to bridge the gap in visualising the nuanced relationship between street greenery design and perceived safety. Furthermore, it offers crucial data-driven support for future urban greenery planning and improving practical guidelines for urban designers, advancing the understanding and application of perceived safety concepts into practice.

2. Literature review

2.1. Impact of the built environment on perceived safety

Poor safety perceptions have been reported to discourage women from active travel (e.g., walking, jogging, biking) (Ding et al., 2020; Graystone et al., 2022; Roman et al., 2024), increasing risks of cardiovascular diseases and threatening mobility justice (Hanson, 2010; Luo et al., 2024). These implications are particularly crucial for women, as they usually perceive a lower degree of safety perception (Cui et al., 2023). Moreover, women showed a 10.7 % lower participation rate in outdoor activities (Outdoor Foundation, 2013) and are more hesitant to visit areas that seem unsafe, which in turn could diminish local businesses (Ceccato & Tcacencu, 2018), suggesting the need to understand what may increase the perceived safety in urban contexts for inclusiveness and equality.

Numerous studies have shown that urban design, especially streetscape design, directly affects the perceived safety of residents and pedestrians (Ewing & Handy, 2009; Lis et al., 2019). Among the determinants of perceived safety, the "natural surveillance" effect in the environment, primarily through street greenery, open sightlines, and building layouts, can significantly enhance perceived safety. For example, the open space between buildings and streets, adequate lighting, and visible street scenes help reduce the concealment of criminal activities, thereby improving residents' sense of security (Holman et al., 2023; Jiang et al., 2018). This concept aligns with Jane Jacobs' "eyes on the street" (Holman et al., 2023; Jacobs, 1961; Jiang et al., 2018), emphasising the role of natural surveillance in enhancing safety by promoting street activity and visibility. It is further supported by the Routine Activity Theory, suggesting that the presence of capable guardians, such as passersby or residents, reduces opportunities for crime (Cohen & Felson, 1979).

2.2. Impact of greenery on perceived safety

Among environmental features, street greenery is key in enhancing perceived safety (Harvey et al., 2015; Kim, 2019). However, the complex relationship between urban greenery and perceived safety could be context-dependent. Crucial factors such as urban form (e.g., urban vs. suburban), degree of enclosure, density, and maintenance conditions all moderate the impact of whether greenery enhances or

Table 1
Comparison of relevant literature.

Category	Title	Target Variables & Data	Key Independent Variables	Model	Findings/Design Relevance	Limitation/Gap
Design Frameworks	Van Veghel et al. (2024)	Perception experience/safety (Place Pulse)	Image features, spatial attributes data	Multinomial Regression	Identifies perception-relevant features; varies by density	No optimisation
Perception Analysis	Porzi et al. (2015)		Image features	Convolutional Neural Networks (CNN)	Identifies visual features linked to safety perception	No visualisation or design integration
	Dubey et al. (2016)				Predicts perception; enables ranking; links to socioeconomic patterns	
	Zhang et al. (2018)			Deep CNN	Reveals perception patterns across cities	
Environmental Studies	Kuo and Sullivan (2001)	Objective safety (crime data)	Vegetation presence, building scale	Ordinary Least Squares (OLS) Regression	Vegetation (−) crime rates	No perception or spatial data considered
	Mouratidis (2019)	Perceived safety (survey + GIS data)	Tree cover, spatial layout		Tree canopy cover (+) perceived safety	No visualisation or design integration
	Troy et al. (2012)	Crime rate (regional statistics)	Tree canopy density, urban gradient		Tree canopy cover (−) crime rates	No design linkage or perception-based
	Kim et al. (2024)	Perceived safety (survey)	Traffic safety features, infrastructure quality		Infrastructure (+) safety perception	No spatial form or design integration

reduces the perceived safety (Donovan & Prestemon, 2012; Jiang et al., 2018; Kuo & Sullivan, 2001; Nasar & Fisher, 1993).

Green plants can enhance the sense of safety by improving visual aesthetics, providing shelter and privacy, reducing anxiety in unfamiliar environments, and offering enclosure, visual softness and cues of maintenance, all of which are consistently associated with perceived safety and social cohesion (Kaplan, 1995; Kuo et al., 1998; Ulrich et al., 1991). Well-maintained and orderly greenery can signal territoriality and care, thus promoting social order and deterring crime (Kuo & Sullivan, 2001). Greenery can improve aesthetics and contribute to safety through environmental comfort (Kondo et al., 2015). In contrast, overly dense and unkempt vegetation might obstruct visual access to surroundings, hampering perceived safety by creating hiding pockets (Jansson et al., 2013; Troy et al., 2016).

Interestingly, when evaluating the safety of parking lots, scenes with higher vegetation coverage are rated as less secure (Shaffer & Anderson, 1985), yet not all greenery obstructs visibility—well-kept lawns, widely spaced high-canopy trees, and low shrubs do not typically provide concealment (Kuo et al., 1998; Michael & Hull, 1994). Furthermore, the density of trees, the diversity of species, and the width of green strips also show varying impacts on perceived safety (Jansson et al., 2013; Jorgensen et al., 2002).

2.3. Measuring perceived safety through ML

In recent years, street view image (SVI) has become an important data source for virtual auditing of urban environments (Hou et al., 2024; Song, Fang, et al., 2024; Zhang et al., 2025), especially concerning perceived safety (Cui et al., 2023; Ye et al., 2025). It visually reflects street characteristics, allowing for more accurate measurement and analysis of pedestrians' safety perceptions in a scalable manner (Zhang et al., 2018). Particularly, digital and image processing technologies provided new opportunities for research on perceived safety using SVIs (Cui et al., 2023; Qiu et al., 2023).

In this regard, the Place Pulse dataset is a crucial open-source dataset measuring and analysing street scene perceptions (Dubey et al., 2016; Salesses et al., 2013). It provides ample (i.e., over 100 thousand) annotated SVIs, which can be used to analyse how different urban environmental features (such as greenery, building design, and road conditions) affect people's perception of safety across diverse urban contexts.

However, most existing studies rely solely on SVI single-source data for analysing perceptions, such as adopting ML to predict perceived safety, and rely on general indicators like the proportion of various urban features in images (Zhang et al., 2018) (see Table 1 for a brief literature comparison). In contrast, this study integrates volumetric environmental data such as tree canopy size, trunk diameter and height from both streetscape imagery and open spatial datasets. This comprehensive approach allows for a deeper understanding of the interactions between volumetric greenery and perceived safety, eventually integrated into the novel data-driven computational urban design workflow.

Notably, we should acknowledge that the mechanism of perceived safety may show divergence across various cultural contexts. Researchers revealed that perceived safety varies across societies, highlighting the need for localised data. Particularly, indicators of social disorder common in Western research, such as public drinking or drug dealing, are less relevant in Chinese cities, where such behaviours are rare and not seen as threats that may influence safety (Liu et al., 2009). Furthermore, graffiti shows similar contrasts. In some Western contexts, it signals disorder and insecurity (James & O'Boyle, 2019), while elsewhere it may be state-sponsored, beautifying, or reinforcing social control (Halsey & Young, 2006). In Asian countries, disorder is instead often linked to noisy, unsupervised youth or neighbourhood quarrels. Perceptions of urban park safety also differ significantly: in the UK and US, fear, especially among women, limits park use (Burgess et al., 1988; Jorgensen et al., 2002; Madge, 1997), whereas in Turkey, parks are seen as safe due to collective use, social cohesion, and visible policing (Özgüner, 2011). This evidence hints that safety perception should be predicted based on locally collected data to derive meaningful implications for urban planning and management.

2.4. Advances in parametric urban design tools

Parametric urban design uses computational algorithms to generate multiple design alternatives based on adjustable parameters, improving efficiency over traditional manual methods (Nagy et al., 2018; Talmor Blaistain & Fisher-Gewirtzman, 2025). By adjusting parameters such as building density, height, and street connectivity, designers can rapidly generate a range of alternatives that integrate environmental, spatial, and social considerations to achieve prospective targets (Koenig et al., 2017; Nagy et al., 2018). Tools such as Rhino and Grasshopper, and plugins like Decoding Spaces, enable the synthesis of street networks, building plots, and volumes with integrated optimisation routines (Vanegas et al., 2012; Çalışkan, 2017). Beyond form-making, parametric tools facilitate performance-driven design processes such as network modelling (Brandes, 2001), economic modelling (Nagy et al., 2018), ecological assessments (Leskens et al., 2017), visibility and perceptual density studies (Batty, 2001; Fisher-Gewirtzman & Wagner, 2003), solar radiation optimisation, daylight availability, and shadow analysis, thus providing a more systematic basis for evidence-based decision-making for planning and designing neighbourhoods (Oxman, 2017; Turrin et al., 2011). These benefits make parametric design particularly relevant for managing the complexities of contemporary urban environments, where diverse spatial qualities and sustainability goals must be reconciled concurrently.

Despite these advances, several gaps remain in the literature. Current parametric frameworks often struggle with scalability when applied to large-scale urban systems and may oversimplify social, cultural, and behavioural dimensions of urban life (Stouffs & Janssen, 2017). While performance metrics such as solar exposure or walkability can be incorporated, the integration of subjective dimensions such as perceived safety, social equity, or cultural identity remains somewhat limited (Koenig et al., 2017). These challenges highlight the need for parametric urban design tools that not only improve computational efficiency but also bridge the gap between quantitative performance measures and qualitative human experience, ensuring design outcomes that are both technically robust and socially meaningful.

2.5. Generative AI in urban design

In this regard, Generative AI (GenAI) including large language models (LLM) has shown great potential in urban planning and design (He et al., 2025; Jiang et al., 2024) throughout the problem formulation, design generation and decision-making full workflow. It can be leveraged to assist planners and designers as a database, a virtual assistant and as an AI-planner team (Fu et al., 2025). It promises better computational efficiency by synthesising realistic spatial patterns and enabling interactive, data-driven workflows (Fernberg & Chamberlain, 2023; Park et al., 2023). Early applications often relied on Generative Adversarial Networks (GANs) for tasks such as generating site plans of cities (Tian, 2021), architectural floor plans (Aalaei et al., 2023; Newton, 2019), street views (Wu et al., 2022), building façades (Meng, 2022), and 3D object models (Wu et al., 2016), demonstrating their excellent capacity to produce diverse urban and architectural forms (Wu et al., 2022). However, training a GAN model demands heavy computation and often struggles with interpretability, limiting its suitability for iterative design workflows (Shen et al., 2020; Zheng et al., 2020).

Recent diffusion-based models, such as Stable Diffusion and ControlNet, allow high-quality, controllable image generation guided by multimodal inputs, including text and structural constraints (Ramesh et al., 2022; Zhang et al., 2023). Applications in urban contexts include generating design options for neighbourhoods (Schlickman & Magana-Leon, 2024), producing landscape visualisations (Awashra et al., 2025; Huang et al., 2024), and converting soundscapes to street views and improving data quality in building footprint generation (Zhou et al., 2024; Zhuang et al., 2024). These developments highlight the growing role of GenAI in enriching visualisation and participatory design processes (Awashra et al., 2025; Fu et al., 2025). However, integrating GenAI with parametric urban design tools for iterative and flexible workflows remains underexplored (He et al., 2025). Notably, prior studies have mainly focused on objective features (Van Veghel et al., 2024) but have largely ignored the integration of perceived qualities in the urban design process. This gap underscores the need to explore context-sensitive, perception-driven GenAI applications in urban design.

3. Data & methods

This section describes the data and methodology, focusing on integrating Place Pulse 2.0 and built environment data to support perception-informed urban design (Fig. 1). The overall workflow involves various tools and software. It begins with a Python-based environment responsible for data preprocessing, statistical modelling, and integration of spatial attributes with image-derived features. Geographic Information System (GIS) software, particularly the open-source software Quantum GIS (QGIS), is employed for spatial identification and analysis (Graser & Olaya, 2015), extracting key built environment variables such as street characteristics. Grasshopper, an integrated visual, node-based programming environment within Rhinoceros 3D, enables dynamic adjustment of spatial parameters to precisely model urban geometry in this study to reflect the urban morphology of the case study site (Costantino et al., 2022). Specifically, the 'Decoding Spaces' plugin within Grasshopper calculates 3D isovists (Koenig et al., 2019), which represent the visible spatial extent from a given viewpoint, effectively simulating and generating images that capture the spatial characteristics of the environment. Finally, the Stable Diffusion algorithm generates high-quality, photorealistic visualisations of optimised design scenarios (Wang et al., 2025). Together, these tools form a looped, perception-driven urban design workflow in which evaluation of perception-based analysis drives design decisions.

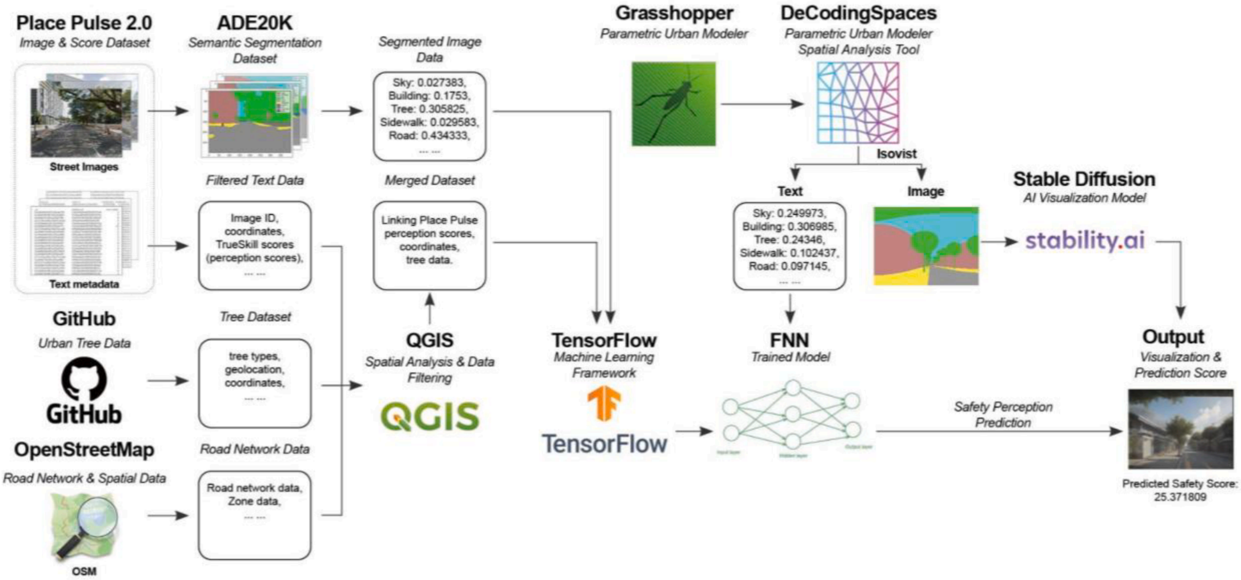


Fig. 1. Workflow of safety prediction and visualisation.

3.1. Case study area

Singapore, a highly urbanised city in Southeast Asia, covers 735.7 km^2 and accommodates a population of 6.04 million as of the end of 2024 (Department of Statistics Singapore, 2024). According to Gallup's Global Safety Report, 94 % of residents reported feeling safe walking alone, placing Singapore second globally and well above the average of around 70 % (Gallup, 2024). This geographical scope supports a focused investigation of how greenery relates to perceived safety within a city that is internationally recognised for its low crime rates and stringent law enforcement (Cheong, 2022; Home Team Behavioural Sciences Centre, 2021). Despite Singapore's consistently high global safety index rankings, safety perception remains spatially uneven. Areas such as Geylang continue to be associated with crime and vice, shaped by historical reasons and social stratification (Greener & Naegler, 2022). This contrast between objective safety and human perception highlights the importance of investigating urban spatial determinants of perceived security to guide planning decisions.

3.2. Uncovering the impact of street greenery characteristics on perceived safety

3.2.1. Data collection

To jointly quantify the impact of street greenery characteristics on perceived safety in urban spaces, this study utilises the Place Pulse 2.0 dataset (Dubey et al., 2016) as the primary source, which comprises 110,988 SVIs annotated with human perceptual rankings on concepts including beauty, liveliness, and safety. These images and the associated data enable the exploration of how urban elements, including trees, roads, and buildings, affect safety perceptions (Zhang et al., 2018). The stated street-view perceptions were gathered through crowdsourcing, with a substantial number of participants globally. It allows for a data-driven analysis that bridges subjective human experiences with objective urban street features, enabling the assessment and optimisation of urban design metrics for enhanced public safety and well-being.

3.2.2. Preprocessing

Several preprocessing steps were conducted to ensure the dataset's relevance and quality. Only those images containing labels of “Safer” perception were retained to focus solely on safety perceptions, and TrueSkill scores were obtained to assess the reliability of human judgments and ensure high-quality data. Furthermore, images depicting isolated buildings or highways were excluded, as they do not provide a comprehensive view of urban environments, where various built elements influence safety perceptions along walking trips. Additionally, this research focused on images containing urban features such as streets, trees, and buildings, which were kept, aligning with the first research objective of examining how greenery and urban characteristics can jointly impact perceived safety. Finally, as this study focuses on the study's geographic context as a proof-of-concept, 2256 images from PP2 taken in Singapore were retained for further analysis.

3.2.3. Constructing ML models

To quantify the physical characteristics of urban elements, image segmentation was applied using the pre-trained PSPNet-50 model. Each image from the Place Pulse 2.0 dataset was segmented into categories such as trees, roads, buildings, and sky. PSPNet-50 was selected for its proven performance in urban scene parsing, having been trained on the ADE20K dataset, a benchmark for dense image segmentation, particularly suitable for urban research (Zhang & Liu, 2021; Zhao et al., 2017). This step provided quantifiable data on the proportion of each urban element, forming a basis for analysing their influences on perceived safety (Zhang et al., 2018).

Notably, although studies have revealed the impact of greenery on perceived safety, their constrained focus on greenery quantity may have led to the mixed findings in prior studies. To overcome this limitation, this study incorporated additional tree-related attributes, including height, girth, species, age and family. These data were obtained from open data platforms, including OpenStreetMap (OSM), as well as Trees.sg, a government platform developed by Singapore's National Parks Board (NParks) that documents detailed information on individual urban trees. Since street view images may not align perfectly with tree locations, QGIS was used to apply a buffer around each image point, associating it with nearby environmental features. This integration ensured a more accurate representation of the visual environment surrounding each location to avoid biases.

The refined dataset was used to train ML models exploring the relationship between environmental features, particularly greenery, and perceived safety. The safety score from the PP2 (Dubey et al., 2016) served as the target variable. Nine ML models were tested to select the best-performing model for subsequent prediction, including traditional regression methods (e.g., OLS), tree-based algorithms (e.g., Random Forest), and deep learning approaches (e.g., CNN). Among them, a Feedforward Neural Network (FNN) was selected for its ability to integrate structured data (e.g., tree attributes, road dimensions) and image-derived features (e.g., segmented element coverage extracted previously from the PSPNet-50 model) (Van Veghel et al., 2024).

In the model training and preprocessing steps, missing values in numerical columns were imputed with the mean, while categorical columns were filled with the most frequent value. A special case was addressed for tree data: if the tree segmentation value was zero, all other tree-related attributes were also set to zero to avoid misleading predictions. We applied feature standardisation and one-hot encoding and engineered an interaction feature, such as tree height multiplied by diameter at breast height, to better capture nonlinear correlations with perceived safety. Splitting dataset into training (80 %) and testing (20 %) sets, the DNN consisted of three fully connected layers with ReLU activation (256, 512, and 256 units), followed by batch normalisation and dropout layers to stabilise training and mitigate overfitting. The final layer was a single linear output node for regression, using the safety score as the targeted output. Our training utilised the Adam optimiser with an initial learning rate of 0.001, along with learning rate decay and early stopping based on validation loss (Fig. 2).

3.3. Integrating environmental characteristics to optimise perceived safety through parametric modelling and simulation

3.3.1. Test scenario selection and design

Urban canyon typologies offer a critical analytical framework for investigating the spatial and perceptual dynamics inherent to high-density environments. This study intentionally transcends site-specific investigations of Singaporean streetscapes, adopting a typology-driven methodology to improve the generalisability of the impact of tree characteristics in urban design. While localised case studies yield valuable contextual insights, their broader applicability is often constrained by unique sociocultural and regulatory contingencies. By abstracting fundamental geometric and urban environmental parameters, such as height-to-width (H/W) and length-to-height (L/H) ratios, previously ignored tree density, and vertical greening coverage, this research systematically examines how variations in canyon morphology influence human-environment interactions. Such parametric abstraction proves particularly salient in Singapore's context, where extreme verticality coexists with low-rise hyper-density within a compact urban fabric.

Two urban canyon typologies (i.e., regular and deep) were chosen to operationalise the parametric framework for urban design. The regular canyon, based on Geylang Lorong, features mid-rise shophouses (4–6 storeys) flanking streets approximately 12 m wide, producing a height-to-width (H/W) ratio of approximately 1, which is typically defined as a regular canyon. The deep canyon, modelled after the Central Business District (CBD), features towers exceeding 30 storeys along streets approximately 25 m wide, producing an H/W ratio greater than 2.0 (Vardoulakis et al., 2003). These typologies reflect morphological extremes within Singapore's urban fabric and embody contrasting socio-spatial conditions. Historically subject to vice regulation and spatial marginalisation, Geylang continues to be perceived as unsafe despite decreasing crime rates (Greener & Naegler, 2022). In contrast, while symbolising financial order, the CBD is increasingly exposed to emergent safety concerns. This spatial juxtaposition provides a critical testing ground for analysing how canyon form and green infrastructure co-produce perceptual outcomes across diverse urban scenarios.

To further support site selection and operationalise the typology-driven methodology, street-level environments across Singapore were preliminarily classified into three categories of perceived safety: low (red), medium (yellow), and high (green), based on Place Pulse scores (Fig. 3). This data-informed classification provides a spatial overview of perception gradients across the city. Interestingly, the two selected canyon typologies, Geylang and the CBD, demonstrate distinct perceptual baselines (Fig. 4). Geylang is primarily associated with low to medium ratings, indicating a need for perceptual enhancement, while the CBD tends to be rated from medium to high, suggesting scope for optimisation. This contrast validates the typology selection by reflecting differing urban conditions and safety perception needs.

3.3.2. Parametric optimisation process

This study further develops a novel data-driven, iterative optimisation framework to refine urban spatial configurations based on perceived safety feedback. Regarding the two distinct urban canyon typologies selected, a new parametric workflow is developed in Grasshopper, integrating spatial data from government portals, Google Earth, and OpenStreetMap. This ensures the models accurately reflect real-world conditions while reducing reliance on extensive empirical data collection.

Previous research has demonstrated that volumetric features of the built environment influence perceived safety, with limited attention to the detailed qualities of urban greenery (Van Veghel et al., 2024). This study focuses on streetscape greenery within urban canyons, with a particular focus on tree-related variables. To align with perceptual design objectives, the parametric model includes only volumetric and spatially operable elements, structured into three adjustable components: buildings, roads, and streetscape greenery. Each component is defined by key spatial parameters such as building height, street width, sidewalk dimensions, tree canopy size, and urban density. Table 2 outlines the adjustable design parameters embedded in the computational workflow. The model enables precise adjustments that align with human perception and urban planning regulations.

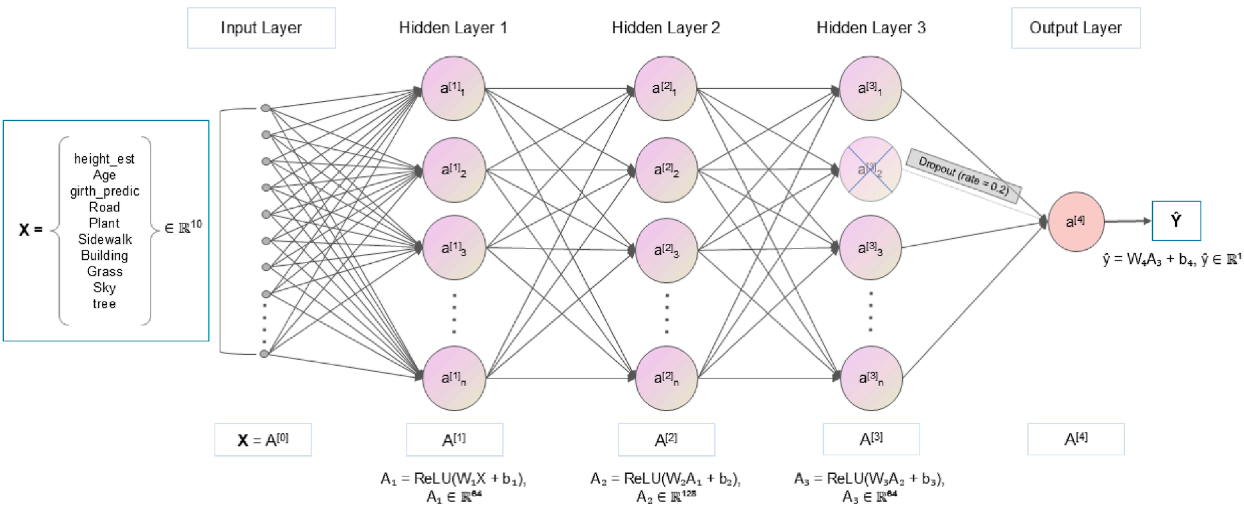


Fig. 2. Customised Feedforward Neural Network (FNN) model architecture.

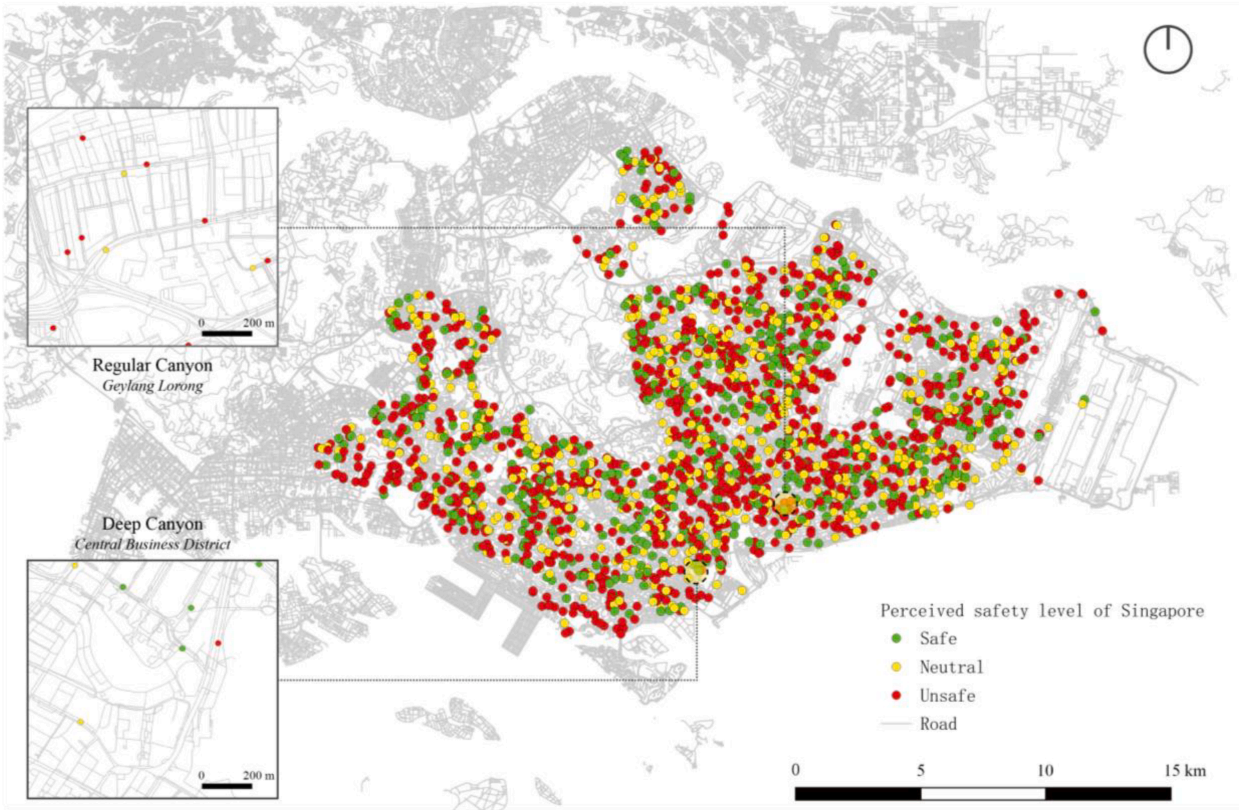


Fig. 3. The visualisation categorises the perceived safety of Singapore streets into three levels: red (unsafe), yellow (neutral), and green (safe). These results guided the selection of prototypes for typology modelling in the subsequent analysis.

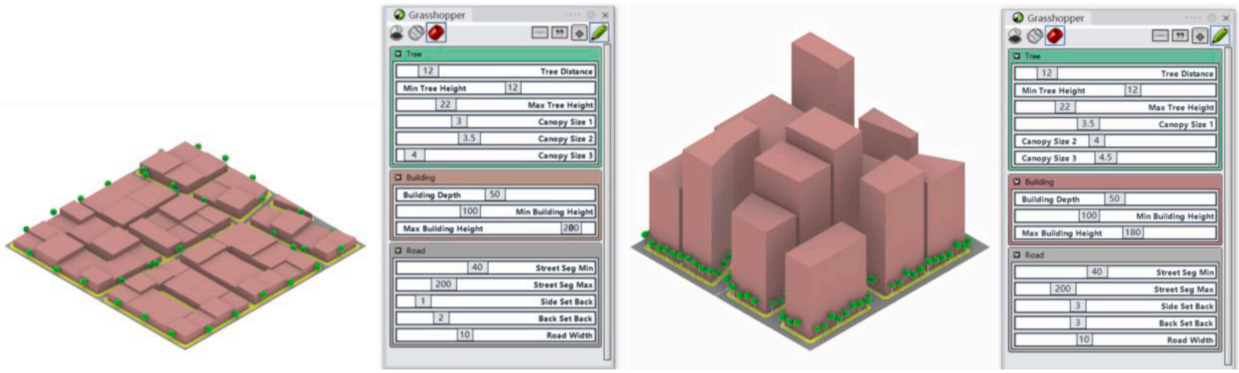


Fig. 4. Adjustable parameters and typology variations in parametric model.

The optimisation process begins with generating a 300 × 300m test model for each canyon type, to maintain consistency and ensure comparability between different area scenarios. This scale captures a representative portion of the urban environment, allowing for a detailed analysis of spatial characteristics while managing computational efficiency. The block size reflects typical urban parcels and provides a meaningful context for assessing human perception in high-density areas.

Isovist analysis is conducted systematically to assess spatial configurations using the ‘DecodingSpaces’ plugin in Grasshopper (https://toolbox.decodingspaces.net). Isovists and isovist fields provide a quantitative means of evaluating how spatial visibility affects human perception, offering insight into the psychological impact of different spatial configurations (Schneider & König, 2012, pp. 355–363). An isovist viewpoint represents the visible area from a specific observation point in a spatial environment. With this plugin, isovist viewpoints of the human eye's perspective can be generated at selected points along the virtual street model. Traditional isovist analysis primarily captures the geometric visibility of space from a specific observation point, focusing on the shape and extent of

Table 2
Adjustable data and input types in parametric modelling.

Categories	Design Variables	Data Type
Tree	Tree Distance	Decimal Number
	Tree Height	Range (Min, Max)
	Trunk Diameter	Decimal Number
	Tree Canopy Size 1, 2, 3 … …	Decimal Number
Road	Street Segment Min	Range (Min, Max)
	Road Set Back	Decimal Number
	Block Side Set Back	Decimal Number
	Block Back Set Back	Decimal Number
	Road Width	Decimal Number
Building	Building Depth	Decimal Number
	Building Height Min	Range (Min, Max)

visible areas while ignoring the semantic content of what is seen (Benedikt, 1979). Although valuable for understanding spatial configuration, this method offers limited insights into how people perceive urban environments. To address this limitation, we integrate isovist analysis with semantic segmentation using PSPNet, pretrained on the ADE20K dataset. This approach retains the spatial visibility mapping of isovist while adding semantic identification of visible elements, enabling a more comprehensive assessment of how spatial and visual factors influence human perception. The segmented isovist outputs were subsequently processed by the trained FNN model in section 3.2.3 to evaluate each scenario, generating a predictive safety score.

Parameter refinement is driven by iterative feedback. Specific spatial elements are adjusted to enhance human perception if the perceived safety score falls below an acceptable threshold. For instance, if an environment is perceived as unsafe, tree spacing is reduced, and canopy coverage is increased to strengthen enclosure and obscure direct sightlines into isolated areas. However, if excessive tree coverage reduces perceived safety, adjustments are made to rebalance security with pedestrian comfort. Similarly, modifications to sidewalk width and street proportions are tested to mitigate discomfort associated with overly narrow urban spaces. Each iteration undergoes re-evaluation through isovist analysis and the ML model for perception prediction, ensuring that spatial adjustments lead to measurable improvements in perceived safety.

This iterative process systematically identifies configurations that optimise urban density while maintaining human-centric spatial

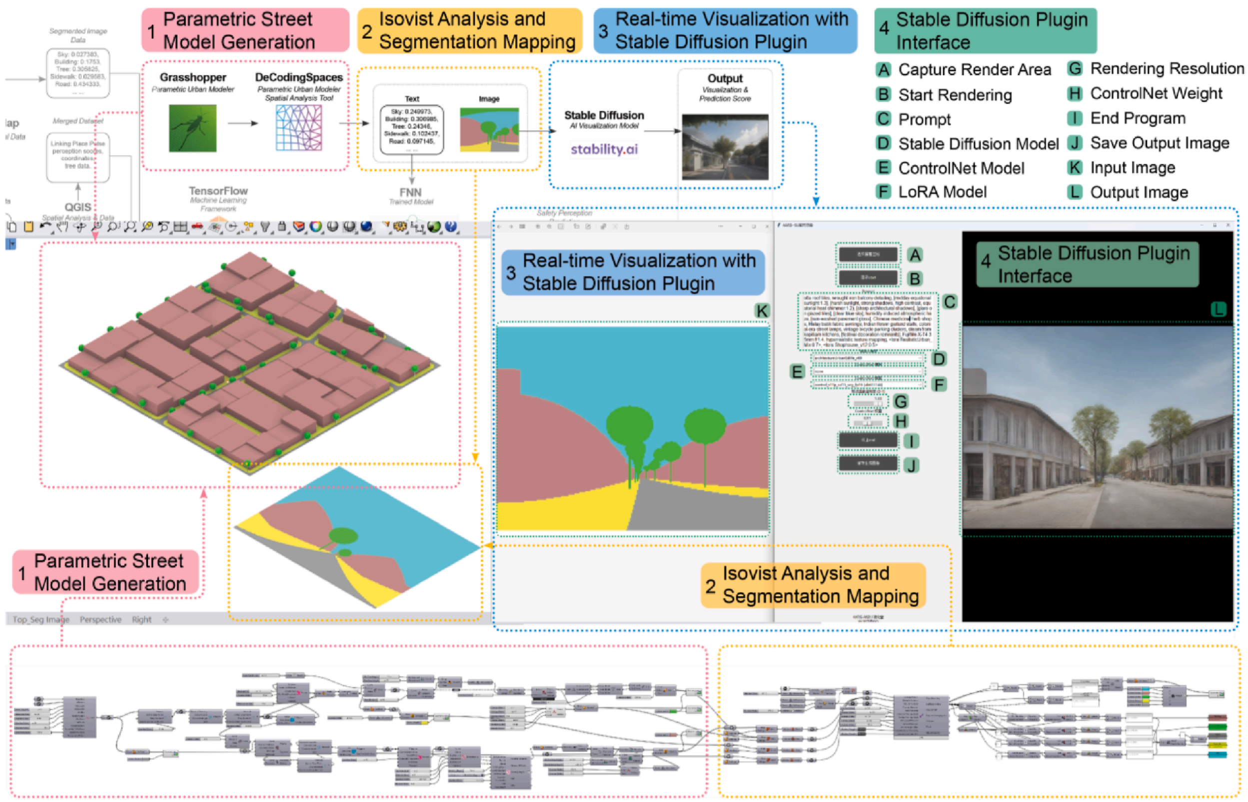


Fig. 5. Parametric-AI workflow for street visualisation.

qualities. By integrating real-time perceptual feedback into parametric modelling, this framework offers a scalable methodology for enhancing high-density urban environments, balancing quantitative optimisation with qualitative spatial experience in urban design workflows.

3.4. Visual simulation of test scenarios

To visually interpret the impact of parametric adjustments for design, the study employs a simulation pipeline that integrates isovist analysis with Stable Diffusion rendering. This approach bridges the gap between abstract parametric data and tangible urban experiences. Isovist 3D generates segmentation images that categorise urban elements into ground, trees, roads, and buildings. These segmentation images are subsequently processed through Stable Diffusion, transforming them into photorealistic street views (Li et al., 2024).

Stable Diffusion, particularly when combined with ControlNet and LoRA, offers precise control over the image generation process. ControlNet is a neural network extension that allows external spatial guidance to preserve structural accuracy in the generated images (Li et al., 2024; Liu et al., 2024; Xu et al., 2024). This capability is essential for visualising complex urban scenarios where architectural styles (e.g., shophouses in regular canyons or modern skyscrapers in deep canyons) and diverse vegetation types must be accurately represented. Unlike traditional stable diffusion web interfaces (such as Stable Diffusion WebUI), this plug-in automatically captures segmentation images and performs instant photorealistic rendering, significantly improving workflow efficiency. The seamless connection between the parametric model output and visual feedback facilitates rapid iteration, making the design exploration process more responsive and intuitive (Fig. 5). More specifically, the module contains four steps:

Step 1. Parametric Street Model Generation: Using parametric modelling techniques in Grasshopper, street environments are systematically generated based on controllable parameters such as building height, street width, and tree layout.

Step 2. Isovist Analysis and Segmentation Mapping: Conducting isovist analysis and creating segmentation images within Grasshopper to extract spatial visibility patterns and categorise key urban elements for AI-driven visualisation.

Step 3. Real-time Visualisation with Stable Diffusion Plugin: Employing the Stable Diffusion plugin for Grasshopper to render realistic street views in real-time based on segmentation inputs, enhancing the intuitive evaluation of urban greenery designs.

Step 4. Stable Diffusion Plugin Functions: The numbered interface explains the functions of the Stable Diffusion plugin, providing a clear overview for real-time visualisation.

The simulation is conducted under different environmental conditions, including various lighting scenarios and atmospheric effects, to capture the full spectrum of human perceptions.

4. Results & discussion

4.1. Impact of greenery characteristics on safety

Of the nine machine-learning algorithms tested, the feed-forward neural network (FNN) delivered the lowest prediction error, with a test-set RMSE of 3.89—about 6 % better than the 2nd-best model (Table 3). While its R^2 may seem modest relative to the 0.20 threshold sometimes cited in social science prediction studies (Van Veghel et al., 2024), such values are common in research on subjective perceptions that are shaped by many unobserved factors. Our model draws exclusively on street-tree metrics and other readily modifiable physical attributes; even in isolation, these variables explain roughly seven percent of the variance in perceived safety, providing a tangible baseline for future work. Notably, perceptions of safety also hinge on social and temporal conditions—pedestrian activity, lighting, surveillance, and neighbourhood reputation (Jiang et al., 2018; Kim et al., 2024; Kuo & Sullivan, 2001; Li et al., 2015; Maas et al., 2009; Van Veghel et al., 2024)—that were intentionally omitted here to isolate the incremental contribution of greenery. Incorporating these additional dimensions falls outside the model's current scope, but offers a clear next step and is likely to improve explanatory power further.

Despite these limitations, Pearson correlation analysis reveals that tree coverage, tree age, tree height, and girth were positively associated with perceived safety, whereas greater sky exposure and wider road area were negatively correlated (Fig. 6). While these

Table 3
Performance benchmark of models.

Model		R2	RMSE
Classical	OLS Linear Regression	0.0403	3.9492
	Lasso Regression	0.0430	3.9435
	Ridge Regression	0.0389	3.9521
Tree-based	Random Forest (RF)	0.0663	3.8954
	Support Vector Regression (SVR)	−0.3283	4.6462
	Extreme Gradient Boosting (XGBoost)	0.0068	4.0176
Deep Learning	CNN	−0.1639	4.3725
	Deep Neural Network (DNN)	0.0125	4.0061
	FNN	0.0712	3.8851

trends align partially with prior urban design studies (Harvey et al., 2015; Lou et al., 2024; Van Veghel et al., 2024), our analysis introduces more granularity by incorporating specific tree attributes, such as age and girth, that have been rarely isolated in perception studies. This level of detail enriches the current understanding of how distinct aspects of streetscape greenery shape perceived safety, suggesting that the presence, maturity, and spatial configuration of trees contribute meaningfully to pedestrian comfort.

4.2. Optimisation results on perceived safety

In both typologies, the selected test scenarios showed improved perceived safety after strategic adjustments to greenery (Table 4), as visually illustrated in Fig. 7. This aligns with previous findings that greenery enhances safety perceptions through natural surveillance and urban environmental comfort (Kondo et al., 2015). The optimisation effectively reduced harsh visibility and improved psychological comfort, reinforcing the importance of environmental factors in urban safety (Fig. 8).

4.2.1. Deep canyon area

In this typology, characterised by narrow street profiles and confined by high-density buildings, the initial scenario featured 15m tree spacing, a canopy size of 3, 3.5, and 4 m, revealing an average safety score of 24.04. Following optimisation (Fig. 9), we increased the canopy size to 3.5, 4, and 4.2 m; accordingly, spacing was reduced to 8m, and trunk size was reduced from 0.2m to 0.15m. The score improved to 24.98, increasing by 3.92 % compared to the existing condition. The enhanced greenery provided both visual and physical shelter, reducing the feeling of vulnerability often associated with narrow, densely built environments. This finding supports Li et al. (2015), who emphasised that greenery could offset psychological discomfort caused by urban canyon effects, enhancing safety through natural concealment and visual relief.

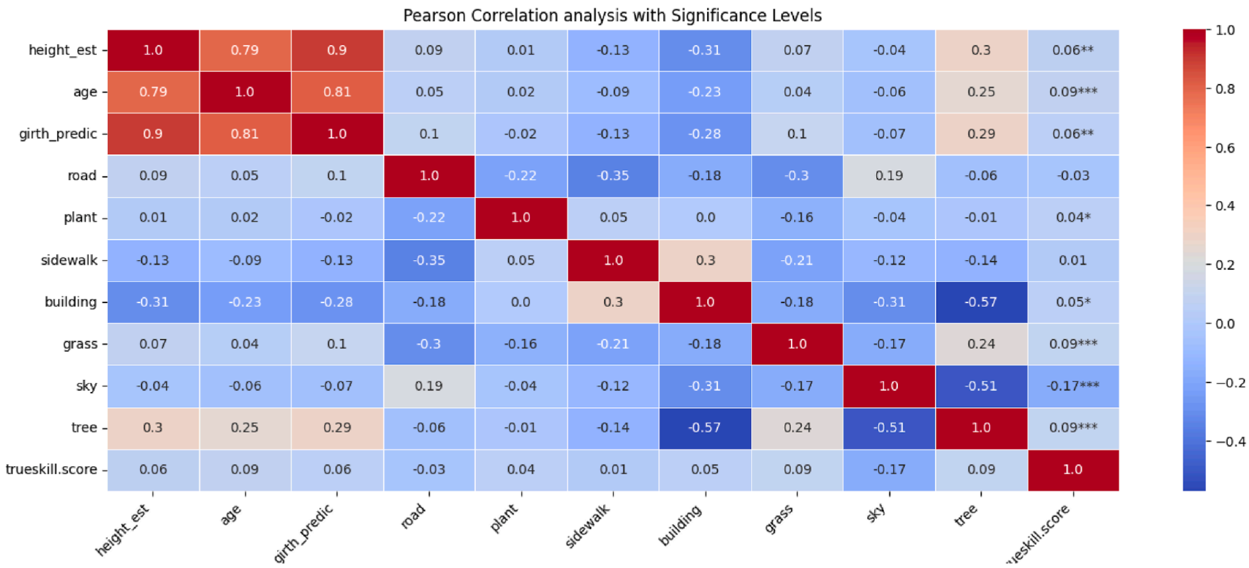
4.2.2. Regular canyon area

In the regular canyon typology, the optimisation addressed the perceived safety concerns arising from increased visibility and openness, despite the area's relatively low building height (Fig. 7). With more open space but lower building height, this typology initially scored 21.54 on perceived safety, using trees spaced 25 m apart, canopy sizes of 3 m, 3.5 m, and 4 m, and a height range of 6–13 m.

After adjusting the canopy size to 3.5 m and 4 m, spacing to 12 m, a tree height range of 7 m–15 m, and keeping the tree girth unchanged, the score increased to 24.74, an improvement by 14.85 %. The intervention provided a better sense of enclosure, aligning with Jacobs' argument (Jacobs, 1961) and Routine Activity Theory (Cohen & Felson, 1979).

4.3. Improved visualisation results

Our visualisation tool offered additional benefits for communicating various scenarios, and the visual simulation framework complements quantitative optimisation by providing a realistic, context-sensitive depiction of urban scenes. By comparing the photorealistic outputs with the perceptual scores derived from the neural network, the study establishes a direct connection between the design interventions and human experience. This integrated approach not only confirms the accuracy of the parametric model but also provides designers and decision-makers with intuitive visual feedback. Specifically, each canyon type was rendered under five



	height_est	age	girth_predic	road	plant	sidewalk	building	grass	sky	tree	trueskill.score
height_est	1.0	0.79	0.9	0.09	0.01	-0.13	-0.31	0.07	-0.04	0.3	0.06**
age	0.79	1.0	0.81	0.05	0.02	-0.09	-0.23	0.04	-0.06	0.25	0.09***
girth_predic	0.9	0.81	1.0	0.1	-0.02	-0.13	-0.28	0.1	-0.07	0.29	0.06**
road	0.09	0.05	0.1	1.0	-0.22	-0.35	-0.18	-0.3	0.19	-0.06	-0.03
plant	0.01	0.02	-0.02	-0.22	1.0	0.05	0.0	-0.16	-0.04	-0.01	0.04*
sidewalk	-0.13	-0.09	-0.13	-0.35	0.05	1.0	0.3	-0.21	-0.12	-0.14	0.01
building	-0.31	-0.23	-0.28	-0.18	0.0	0.3	1.0	-0.18	-0.31	-0.57	0.05*
grass	0.07	0.04	0.1	-0.3	-0.16	-0.21	-0.18	1.0	-0.17	0.24	0.09***
sky	-0.04	-0.06	-0.07	0.19	-0.04	-0.12	-0.31	-0.17	1.0	-0.51	-0.17***
tree	0.3	0.25	0.29	-0.06	-0.01	-0.14	-0.57	0.24	-0.51	1.0	0.09***
trueskill.score	0.06	0.09	0.06	-0.03	0.04	0.01	0.05	0.09	-0.17	0.09	1.0

Fig. 6. Pearson Correlation analysis.

Table 4
Pre- and post-optimisation for test scenarios and parameter adjustments.

Typology	Avg. Pre-Optimisation Score	Tree Canopy Size Range (m)	Tree Height Range (m)	Trunk Size (m)	Tree Distance (m)	Avg. Post-Optimisation Score	Change in Safety (%)
Deep canyon	24.04	3, 3.5, 4 → 3.5, 4, 4.2	12- 22 → 12- 22	0.2 → 0.15	15 → 8	24.98	+3.92
Regular canyon	21.54	3, 3.5, 4 → 3.5, 4	6- 13 → 7- 15	0.2 → 0.2	25 → 12	24.74	+14.85

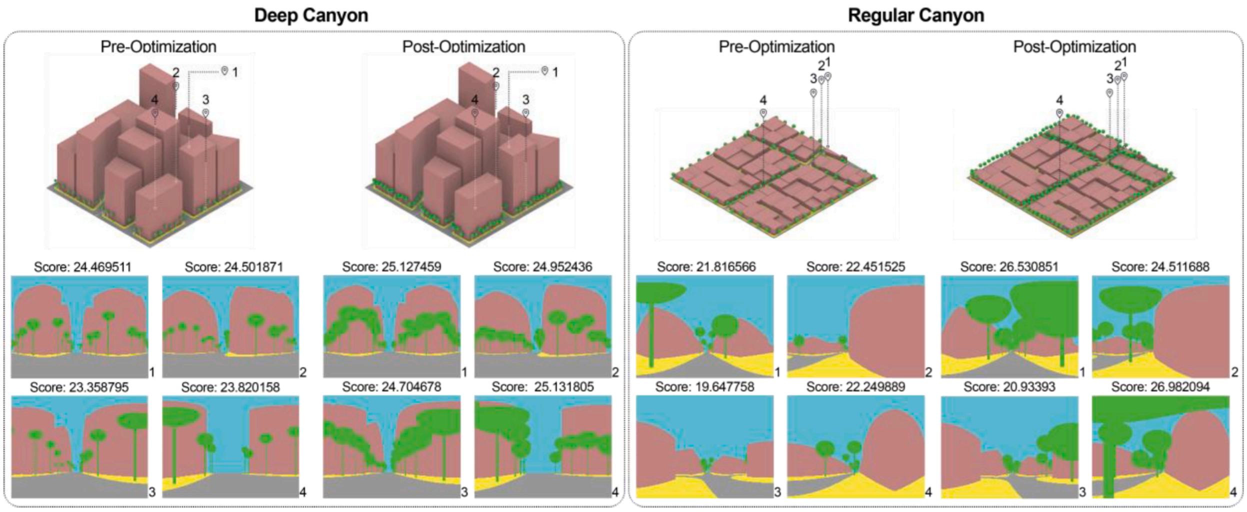


Fig. 7. Perceived safety score prediction (Before and after optimisation for two types).

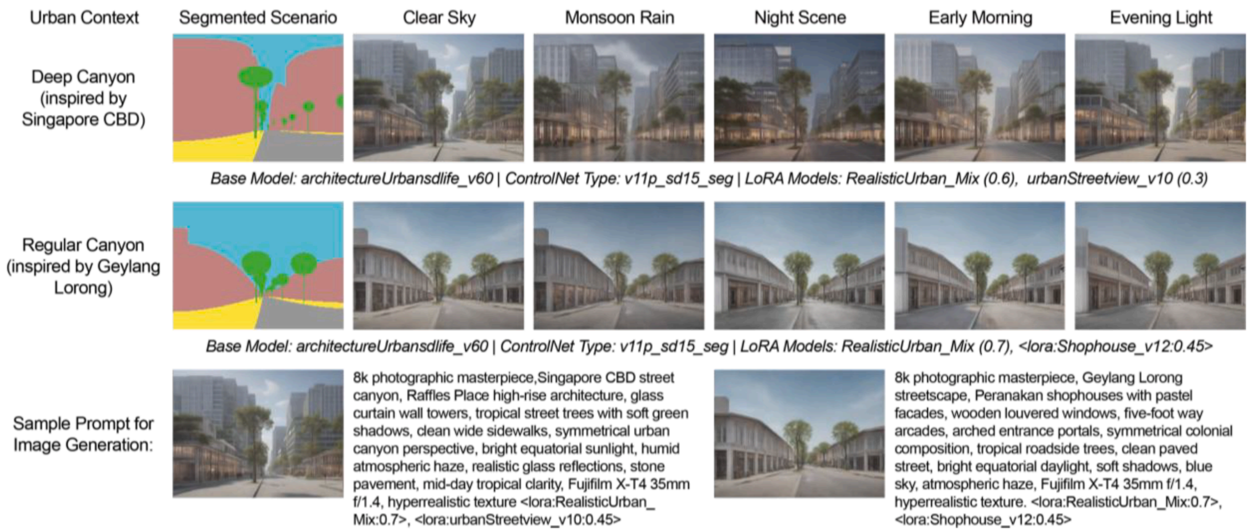


Fig. 8. Examples of visualisation simulations in varying weather, environmental, and lighting conditions.

different environmental conditions: clear sky, monsoon rain, nighttime, early morning, and evening light (Fig. 8).

This proposed workflow can be directly embedded into early-stage urban design by linking parametric modelling with real-time perception evaluation. Within Grasshopper, designers adjust variables such as building height, street width, and greenery density while receiving immediate feedback on predicted safety perception. This enables rapid iteration, supports data-informed decisions, and allows flexible adaptation to different urban contexts without relying on high-resolution datasets. The system thus facilitates context-sensitive spatial strategies that align urban form with human experience and psychological well-being.

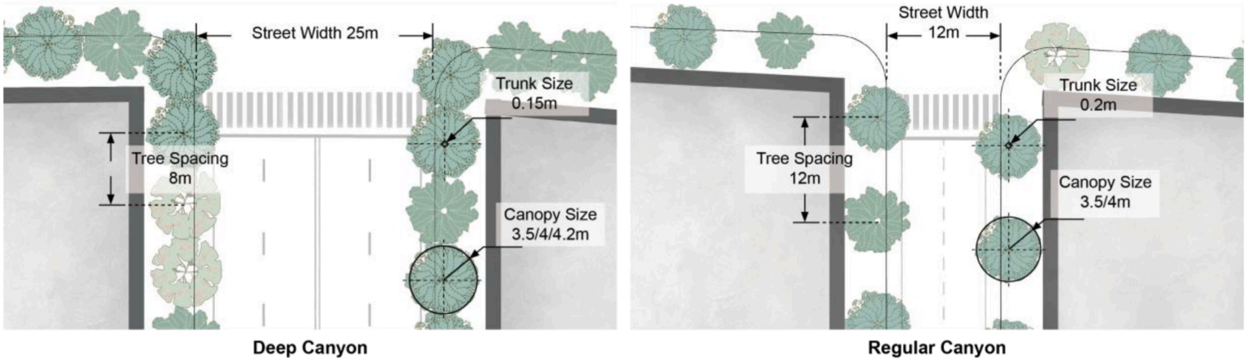


Fig. 9. Optimal streetscape configurations for perceived safety for both typologies.

4.4. Planning implications

This AI-powered workflow suggests that urban form mediates the effectiveness of greening strategies on perceived safety. While deep canyon areas showed limited improvement (+3.92%), regular canyon streets achieved significantly greater gains (+14.85%), implying that the latter may offer higher return on investment. In deep canyon areas where vegetation may already be near saturation, it is worth exploring alternative strategies such as reconfiguring pedestrian zones, introducing central green strips, or adjusting street-wall proportions to achieve additional benefits. Future applications could integrate such qualitative attributes into parametric models, supporting more targeted, perception-informed urban design.

4.5. Limitations and future work

However, the study also contains several limitations related to data granularity, particularly regarding tree canopy dimensions. First, while the workflow reduces dependence on high-precision datasets, more comprehensive urban greenery data, such as classified canopy shape, seasonal variation, tree species, and detailed three-dimensional greenery data, would enhance model accuracy and applicability. Second, although generative models like ControlNet and LoRA are helpful for visualising design interventions, they still struggle with localised architectural accuracy, limiting their utility in culturally specific contexts. This shortcoming highlights the limitations of the current generative model's performance and suggests the need to improve the training dataset to cover more diverse environmental scenarios, especially for culture-specific architectural styles, using self-customised data for a domain-specific fine-tuned model. Finally, the regular canyon type inspired by Singapore's Geylang Lane exhibits poor visual quality in certain weather conditions, such as at night or during rainy and overcast periods. This difference is likely due to the limitations of the corresponding LoRA stylisation model, which may not be adequately trained on different weather and lighting datasets. Future work should focus on developing more efficient fine-tuning methods for both LoRA and ControlNet to improve model adaptability across diverse weather and lighting conditions (Hu et al., 2021). Enhancing real-time Stable Diffusion plugins with expanded parameter controls will further streamline workflow integration and usability for urban design applications.

5. Conclusion

As high-density cities proliferate, safeguarding psychological wellbeing—especially perceived safety—must become an explicit design objective. Yet prevailing practice still treats human perception as an afterthought, relying on static heuristics rather than empirical evidence. We introduce an end-to-end computational workflow that couples parametric typology modelling, ML–based perception prediction and real-time generative visualisation. Designers can vary street morphology or vegetation and receive instant, data-driven feedback on how those changes are likely to be felt at the human scale.

This research introduces a novel design workflow to improve the existing urban design method, integrating parametric urban typology modelling, ML-based perception prediction, and real-time AI-assisted generative visualisation. Beyond validating the general benefits of greenery, this study first identifies specific vegetation characteristics that influence safety perception. Variables such as canopy size, tree height, and tree spacing were found to significantly affect perceived safety and can be strategically adjusted to enhance the psychological effectiveness of green infrastructure in the trained ML model. Informed by these findings, designers can subsequently simulate variations in urban form, such as building height, greenery distribution, and street width, and receive immediate visual and data-driven insights into how these changes affect human perception.

Applying this workflow to two representative urban typologies, the deep and regular canyons, revealed that strategically optimising street greenery significantly enhances perceived safety. In the deep canyon, increasing tree canopy size effectively reduces excessive openness and harsh visibility. In regular canyon settings, dense and well-positioned greenery creates sheltering effects, fostering comfort and a sense of security. However, the findings also suggest that greenery alone may not significantly enhance safety perception in current highly developed districts. In these cases, modifying street composition, such as introducing central medians or reducing the scale of vehicular lanes, may pose a more notable impact.

In summary, this study presents a novel computational design workflow that effectively diagnoses and improves perceived safety in urban settings. By integrating ML models with local urban form and perception data, the approach enables human-centred design decisions and generates insights that are transferable across high-density urban contexts. The proposed workflow is easily parameterised, allowing application at multiple spatial scales, from individual streets to entire cities, and can be adapted to diverse geographic, cultural, and policy settings. This adaptability allows the workflow to be scaled and applied in other cities, enabling planners to simulate local urban forms, identify key perception-related spatial variables, and develop evidence-based design interventions tailored to their specific needs. It shows that when strategically integrated, advanced computational tools can bridge urban analytics and urban design more closely with human needs, fostering safer, more inclusive, and emotionally responsive city environments.

CRediT authorship contribution statement

Xinyu Tan: Writing – original draft, Visualization, Validation, Investigation, Data curation, Conceptualization. **Qiwei Song:** Writing – original draft, Supervision, Project administration, Investigation, Conceptualization. **Xun Liu:** Writing – review & editing, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Waishan Qiu:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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